Department of Computer Science and Engineering

UIET, CSJM University, Kanpur

C Programming Lab Assignment

Decision Making using `if`, `if-else`, and Nested `if`

Student Name:	**Roll Number:**
Branch / Section:	**Date:**

Laboratory Instructions

Instructions

1. Create a separate directory for this assignment.
2. Save each program using the filename suggested in the question.
3. Read the complete problem statement before writing the program.
4. Test every boundary value mentioned in the problem.
5. Use only the concepts taught in class: variables, input/output, arithmetic operators, relational operators, logical operators, and `if-else`.
6. Do not use loops, arrays, functions, or `switch` statements.

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Problem 1: Examination Entry

Story

At the UIET examination gate, a computer checks a student's attendance before allowing entry. A student needs at least 75 percent attendance to appear in the examination.

Write a C program that reads attendance percentage and prints whether the student may appear in the examination.

Input

One real number A, representing attendance percentage.

Output

Print exactly one message:

```
ELIGIBLE FOR EXAMINATION
```

or

```
ATTENDANCE SHORTAGE
```

Constraints

$$0.0 \leq A \leq 100.0$$

Sample Test Cases

Input	Output
75	`ELIGIBLE FOR EXAMINATION`
74.9	`ATTENDANCE SHORTAGE`

Suggested filename: **`exam_entry.c`**

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Problem 2: Temperature Monitor

Story

A small weather-monitoring program installed in the UIET laboratory reads an integer temperature. The program must identify whether the temperature is above zero, below zero, or exactly zero.

Write a C program that reads one integer temperature and prints its type.

Input

One integer T, representing temperature.

Output

Print exactly one message:

`POSITIVE TEMPERATURE`

or

`NEGATIVE TEMPERATURE`

or

`ZERO TEMPERATURE`

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Constraints

$$-1000 \leq T \leq 1000$$

Sample Test Cases

Input	Output
-8	`NEGATIVE TEMPERATURE`
0	`ZERO TEMPERATURE`
27	`POSITIVE TEMPERATURE`

Suggested filename: `temperature_monitor.c`

Problem 3: Internal Marks Validator

Story

Before storing internal marks in the department record, the UIET computer must verify that the entered marks are valid. Internal marks are out of 30, so a value below 0 or above 30 must not be accepted.

Write a C program that reads internal marks and checks whether the value is valid.

Input

One real number M, representing internal marks.

Output

Print exactly one message:

```
VALID MARKS
```

or

```
INVALID MARKS
```

Constraints

$$-100.0 \leq M \leq 100.0$$

Requirement

Use the logical AND operator **`&&`**.

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Sample Test Cases

Input	Output
0	`VALID MARKS`
30	`VALID MARKS`
30.1	`INVALID MARKS`
-1	`INVALID MARKS`

Suggested filename: **`marks_validator.c`**

Problem 4: Lab Computer Access

Story

The Unix laboratory has a simple access system. A student may use a computer only when the entered access code is either 101 or 202. All other codes must be rejected.

Write a C program that reads an access code and prints whether access is granted.

Input

One integer C, representing the access code.

Output

Print exactly one message:

```
ACCESS GRANTED
```

or

```
INVALID ACCESS CODE
```

Constraints

$$0 \leq C \leq 999$$

Requirement

Use the logical OR operator `||`.

Sample Test Cases

Input	Output
101	`ACCESS GRANTED`
202	`ACCESS GRANTED`
102	`INVALID ACCESS CODE`

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Suggested filename: `lab_access.c`

Problem 5: Two-Number Announcement

Story

During a programming activity, two students enter their roll numbers into a computer. The computer must announce which number is larger. If both numbers are the same, it must clearly report equality.

Write a C program that reads two integers and prints the correct comparison.

Input

Two integers A and B.

Output

Print exactly one message:

```
FIRST NUMBER IS LARGER
```

or

```
SECOND NUMBER IS LARGER
```

or

```
BOTH NUMBERS ARE EQUAL
```

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Constraints

$$-100000 \leq A, B \leq 100000$$

Sample Test Cases

Input A	Input B	Output
18	21	`SECOND NUMBER IS LARGER`
45	12	`FIRST NUMBER IS LARGER`
7	7	`BOTH NUMBERS ARE EQUAL`

Suggested filename: `compare_numbers.c`

Problem 6: The UIET Scholarship Desk

Story

At the UIET scholarship desk, an employee enters a student's percentage marks and attendance percentage into a computer.
The computer must first reject incorrect data. A value below 0 or above 100 is invalid.
A student who has not passed cannot receive any scholarship, even if the attendance is excellent.
For students who have passed, the scholarship desk follows these rules:

- A student with attendance below 65% is detained.
- A student with attendance from 65% to less than 75% receives a Conditional Scholarship only when marks are at least 90.
- A student with attendance of at least 75% receives:
  - Full Scholarship for marks at least 85;
  - Half Scholarship for marks from 70 to less than 85;
  - only Pass status for marks below 70.

Write a program that prints the final decision of the scholarship desk. Use nested **`if-else`** statements only. Do not use loops or an **`else if`** ladder.

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Input

Two real numbers:

- M: percentage marks;
- A: attendance percentage.

Output

Print exactly one of the following messages:

```
INVALID INPUT
FAIL
FULL SCHOLARSHIP
HALF SCHOLARSHIP
CONDITIONAL SCHOLARSHIP
PASS
LOW ATTENDANCE
DETAINED
```

Input Constraints

The entered values may lie in the following range:

$$-100.0 \leq M, A \leq 200.0$$

However, valid marks and attendance must each lie from 0 to 100. For every value outside this valid range, print `INVALID INPUT`.

Sample Test Cases

Marks	Attendance	Output
86	75	`FULL SCHOLARSHIP`
70	75	`HALF SCHOLARSHIP`
69.9	75	`PASS`
90	74.9	`CONDITIONAL SCHOLARSHIP`
89.9	74.9	`LOW ATTENDANCE`
80	64.9	`DETAINED`
39.9	100	`FAIL`
101	80	`INVALID INPUT`

Challenge Test Cases

Before running the program, predict the output for:

```
40 65    70 74.999    85 75    90 65    0 0    -1 80
```

Suggested filename: `scholarship_desk.c`

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Remember: A C program may compile correctly and still make an incorrect decision. Test the boundary values and trace every condition.